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High Level Programming Extensions – Triggers

Program 1 :

Create a trigger in MySQL to log the changes of the STUDENT table with fields ID, Name and Email. Also create a new table named EMPLOYEES_AUDIT to keep the changes of the employee table. Create a BEFORE UPDATE trigger that is invoked before a change is made to the employees table.

```
mysql> create database collegedb;
Query OK, 1 row affected (0.06 sec)

mysql> use collegedb;
Database changed
mysql> create table employees(id int primary key,name varchar(100),email varchar(100));
Query OK, 0 rows affected (0.03 sec)

mysql> insert into employees values(1,'sonu','sonu@gmail.com'),(2,'pranathi','pranathi@gmail.com'),(3,'roopa','roopa@gmail.com');
Query OK, 3 rows affected (0.06 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from employees;
+----+-----+-----+
| id | name  | email          |
+----+-----+-----+
| 1  | sonu  | sonu@gmail.com |
| 2  | pranathi | pranathi@gmail.com |
| 3  | roopa | roopa@gmail.com |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql> create table employees_audit(id int,name varchar(100),email varchar(100),changed_at timestamp default current_timestamp);
Query OK, 0 rows affected (0.17 sec)

mysql> desc employees_audit;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int  | YES  |     | NULL    |       |
| name  | varchar(100) | YES  |     | NULL    |       |
| email | varchar(100) | YES  |     | NULL    |       |
| changed_at | timestamp | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> delimiter //
mysql> create trigger before_employee_update before update on employees for each row begin insert into employees_audit(id,name,email) values(old.id,old.name,old.email);end //
Query OK, 0 rows affected (0.06 sec)

mysql> delimiter ;
mysql> update employees set email='balikana@gmail.com' where id=1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from employees;
+----+-----+-----+
| id | name  | email          |
+----+-----+-----+
| 1  | sonu  | balikana@gmail.com |
| 2  | pranathi | pranathi@gmail.com |
| 3  | roopa | roopa@gmail.com |
+----+-----+-----+
```

```
mysql> delimiter //
mysql> create trigger before_employee_update before update on employees for each row begin insert into employees_audit(id,name,email) values(old.id,old.name,old.email);end //
Query OK, 0 rows affected (0.06 sec)

mysql> delimiter ;
mysql> update employees set email='balikana@gmail.com' where id=1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from employees;
+----+-----+-----+
| id | name  | email          |
+----+-----+-----+
| 1  | sonu  | balikana@gmail.com |
| 2  | pranathi | pranathi@gmail.com |
| 3  | roopa | roopa@gmail.com |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from employees_audit;
+----+-----+-----+-----+
| id | name  | email          | changed_at |
+----+-----+-----+-----+
| 1  | sonu  | sonu@gmail.com | 2026-08-20 11:19:54 |
+----+-----+-----+-----+
1 row in set (0.00 sec)
```